

Xiaohang Ji

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Professional Summary

Software developer with strong experience in **Java backend development**, **distributed systems**, and **full-stack applications**. Skilled in building **scalable**, **secure**, and **production-ready systems** using **Spring Boot**, **microservices architecture**, and modern frontend technologies. Experienced in **API design**, **system architecture**, and **cloud deployment**, with hands-on projects involving **AI systems**, **gateway security**, and **real-time streaming**.

Project

Ji-AI-Agent — Multi-Agent AI System (Production Deployment) [GitHub](#) [Demo](#)

Technologies: [Java](#) [Spring Boot](#) [Spring AI](#) [React](#) [PostgreSQL](#) [pgvector](#) [Docker](#) [Nginx](#)

Key Achievements:

- Hands-on experience implementing **RAG pipelines** with **pgvector** and custom retrieval strategies, including **query rewriting** and vector-based semantic search
- Familiar with **MCP (Model Context Protocol)** and **SSE streaming**, with the ability to develop and deploy **tool-integrated AI agents** and scalable backend services
- Proficient in **Java 21** and **Spring Boot**, with experience using tools such as **Hutool**, **Lombok**, and **Knife4j** to improve development efficiency and maintain code quality
- Experienced in building **AI-powered applications** using frameworks like **Spring AI** and **LangChain4j**, with solid understanding of **RAG**, **tool calling**, **ReAct**, and **CoT**

API Interface Open Platform [GitHub](#) [Demo](#)

Technologies: [Spring Boot](#) [Spring Cloud Gateway](#) [Docker](#) [Swagger + Knife4j](#) [Dubbo](#) [MyBatis-Plus](#)

Key Achievements:

- Designed a **microservice-based API platform** with modular architecture (web, gateway, SDK, common) using **Maven multi-module management**
- Implemented a **secure API authentication system** using **AK/SK + HMAC-SHA256**, with **timestamp** and **nonce validation** for replay protection
- Developed a **Spring Boot Starter SDK** enabling **one-line API invocation**, significantly improving developer integration efficiency
- Built a centralized **API gateway** with **Spring Cloud Gateway** to handle **routing**, **authentication**, **rate limiting**, and **usage tracking**

Professional Experience

IT Student - *Agriculture and Agri-Food Canada*

2022.09 - 2022.12/2023.09 - 2024.09

- Conducted **software and hardware testing**, identifying and resolving **critical bugs** before production release
- Analyzed **backend workflows** and **database queries**, improving **system performance by 15%**
- Maintained the **inventory system**, improving **data accuracy by 20%**, and collaborated within an **Agile team**

Education

Algonquin College – Computer Engineering Technology (Computing Science)

Sep 2021 – Sep 2024 | GPA: 3.8 / 4.0